

PhD Research Proposal

Using Large Language Model-Powered Multi-Agent Simulation to Model Policy Implementation and Policy Evaluation

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Abstract

This thesis asks whether large language model (LLM) powered multi-agent simulations can serve as a valid methodological complement to existing policy pre-implementation planning and evaluation approaches. Recent advances in generative agent-based modelling make it possible to create large populations of individually distinct agents that autonomously interact within a specified policy environment. This opens two possibilities. First, simulating policy implementation in controlled settings to explore how different stakeholder groups might respond under varying conditions. Second—and less explored—simulating policy evaluations themselves: running synthetic RCTs, quasi-experiments, or qualitative case studies to stress-test research designs before committing real-world resources. For these capabilities to be trusted in government settings, however, they need to be developed in a systematic and well-studied manner. At a minimum, the models powering these simulations should be fine-tuned on the demographics, institutional norms, and cultural context of the populations on which policies will be implemented, to increase real-world applicability.

I propose to do this empirically using well-documented Singapore education policy cases, with an Australian comparator to test cross-contextual applicability. I also develop a locally fine-tuned simulation model for Singapore's governance context, released as an open-source resource. This thesis has three practical deliverables: a methodology for using LLM simulation to inform policy evaluation design; a framework positioning simulation within evaluation approaches and pre-implementation evaluation tools; and governance guidelines for its responsible use by government for policy decision making.

Aims and Significance

This thesis aims to investigate the potential — and the practical and ethical boundaries — of using LLM-powered multi-agent simulation to model policy implementation and policy evaluation before real-world rollout. It develops and empirically tests the use of generative agent-based modelling to explore how policies might be implemented and evaluated in advance and produces a framework for its responsible integration into public policy and administration practice.

This matters for three reasons. First, current pre-implementation policy and evaluation tools remain static and unable to qualitatively model emergence, discretion, or resistance. Second, amidst rapid advances in generative AI, we need parallel studies that validate these capabilities against known policy outcomes and develop governance guidelines for their use in government. Formal research will always lag the technology — but the work still needs to be done, because subsequent research will build on it. Third, no fine-tuned simulation model exists for non-Western governance contexts as of writing. This thesis aims to address all three gaps by producing a sound methodology for policy simulation, a framework for positioning evaluation simulations, and practitioner-facing governance guidelines — alongside the academic contribution. The empirical focus on Singapore education policy, with an Australian comparator, ensures relevance to the Asia-Pacific region and to Crawford POGO's research community.

Research Questions

The primary research question is *“To what extent can LLM-powered multi-agent simulation function as a valid methodological complement to existing pre-implementation policy simulations and evaluation approaches?”*

The research sub-questions are below. Sub-questions 1–2 are empirical, 3–4 are methodological, and 5 addresses governance and ethics.

1. To what extent can LLM agents—using publicly available policy documents as seed material—reproduce known patterns of ground-level interpretation, actions and implementation resistance in past Singapore and Australian policy rollouts?
2. How does a locally fine-tuned simulation model compare to generic LLM baselines in cultural validity and behavioural fidelity for selected policy contexts?
3. Under what validity conditions can anticipatory simulation outputs be trusted as inputs to policy design and evaluation design decisions?
4. How should LLM-based simulation be integrated into existing evaluation approaches (e.g., realist evaluation, developmental evaluation) and evaluation tools (e.g., theory of change, logic models)?
5. What governance guidelines and good practices—relating to bias, accountability, interpretability, and responsible use—are necessary for LLM-based simulation to be deployed ethically by governments?

Literature Review

National policies and implementation variation. National policies are often designed with broad outcomes and the expectation of consistent implementation across diverse contexts. However, the realities of local interpretation, resource constraints, and institutional sense-making frequently result in significant variation in how policies are enacted in practice. Lipsky's (2010) foundational account of street-level bureaucracy established that frontline

actors — teachers, social workers, administrators — are not passive conduits for policy but active agents who reshape it through discretion. In education specifically, Spillane’s (2006) distributed leadership perspective shows how school actors interpret policy through their own cognitive frames and institutional roles, while Trinidad (2024) maps the intra-school structural, network, and ecological processes that mediate between system-level directives and classroom practice. This is relevant for Singapore’s context as though Singapore’s education policy implementation has been well-documented at the system level (Gopinathan & Lim, 2021; Viennet & Pont, 2017) it is far less so at the school level where schools have significant autonomy to customise policy implementation for local needs (Chong & Lee, 2021). We have existing frameworks to describe and predict educational policy implementation that could be adopted for simulation — but current pre-implementation tools cannot do this.

Classical ABM and its limits. Agent-based modelling has been applied to policy questions for over two decades, spanning urban planning, public health, and disaster management (Axelrod, 1997; Epstein, 2006). Its core insight — that complex macro-level patterns emerge from interactions among heterogeneous, locally-acting agents — makes it a natural fit for studying distributed implementation processes. But classical ABM relies on quantitative rule-based agents whose behaviour is specified in advance. These agents cannot reason, argue, or exercise contextual judgement. They cannot model the linguistic and social dynamics of a parent-teacher meeting or the cultural norms shaping how a school principal interprets education policies.

The generative agent-based modelling turn. Park, O’Brien, Cai, Morris, Liang, and Bernstein’s (2023) ‘Generative Agents’ experiment showed that LLM-powered agents could produce emergent social behaviours — forming relationships, coordinating activities, exhibiting plausible individual variation — without explicit rules. Gao et al. (2024) surveyed the expanding landscape of LLM-empowered ABM while Ghaffarzadegan, Majumdar, Williams, and Hossenichimeh (2024) formalised this as Generative Agent-Based Modelling (GABM) establishing the field’s conceptual foundations. On the infrastructure side, there have been significant recent advances: CAMEL-AI’s OASIS engine scales LLM agent interactions to over one million exchanges; MIT’s AgentTorch introduced LLM archetypes for population-scale simulation; and MiroFish in 2026 operationalised a full pipeline accepting data and documents as seed inputs. For the first time, it is technically feasible to simulate how individual teachers, parents, or school principals might receive, interpret, and respond to a proposed policy before it is implemented. This is a qualitative shift in the kinds of questions pre-implementation analysis can address. Larooij & Tornberg (2026) argued that validation remains the central unsolved challenge: LLM agents may exacerbate existing ABM validity concerns through black-box reasoning, cultural biases, and stochastic outputs. I treat this validation challenge not as a limitation to manage but as the core research problem.

LLM simulation applied to policy. The most recent published study is Li, Das, and Shirado’s (2025) year-long engagement with emergency preparedness policymakers, where 13,000 LLM agents simulated community crisis responses. Their three design implications — start with verifiable scenarios, use simulation to elicit tacit knowledge, treat simulation and policy as co-evolving — are instructive. But their contribution addresses the design and trust question in a physical crisis domain, while this thesis addresses the evaluation question in a social and educational policy domain.

Simulating policy evaluations. This thesis also investigates the potential to simulate policy evaluations. If we can model how a policy plays out, we should also be able to simulate evaluation designs such as RCTs, quasi-experimental studies, and longitudinal studies in the

same synthetic environment, identifying potential flaws and blind spots before committing real-world resources. Taken further, we could empirically test how closely simulated evaluation findings track actual outcomes, informing decisions about where real-world evaluation effort is most needed and where simulation can supplement it. This goes beyond conventional scenario planning exercises, which typically ask stakeholders to react to hypothetical findings in a facilitated setting — here, the scenarios are generated computationally from the ground up, with heterogeneous agents responding to a variety of policy conditions. In doing so, this thesis aims to contribute to the evaluation field by expanding and complementing current evaluation theories such as developmental evaluation (Patton, 2010) and pre-implementation evaluation tools such as theories of change (Funnell & Rogers, 2011) and logic models. No study has yet explored this possibility in a systematic way.

Fine-tuning and local grounding of LLM models. Current foundation models are predominantly trained on Western, English-language corpora. Research on cultural bias (Kovač, Sawayama, Portelas, Colas, Dominey, & Oudeyer, 2023; Santurkar, Durmus, Ladhak, Lee, Liang, & Hashimoto, 2023) shows this produces systematically skewed agent behaviour in non-Western contexts. For a simulated Singaporean teacher to respond with cultural fidelity to an MOE directive, the underlying model needs knowledge of Singapore’s educational norms, institutional culture, and social expectations. Fine-tuning via low-rank adaptation (LoRA; Hu et al., 2021) offers a computationally efficient path to local grounding, while retrieval-augmented generation (RAG) provides a complementary approach. No fine-tuned simulation model currently exists for Singapore or Southeast Asian governance contexts. The methodology developed here is designed for replication by any jurisdiction wishing to localise simulation for its own policy environment.

AI governance within governments. Singapore recently developed the Model AI Governance Framework for Generative AI (IMDA & AI Verify Foundation, 2024) and the Model AI Governance Framework for Agentic AI (IMDA, 2026). The agentic framework addresses risk assessment, human oversight, technical controls, and end-user responsibility for multi-agent systems — all directly relevant to the simulation tools this thesis employs. However, these frameworks govern the development and deployment of AI products and services for private end-users and the public. They do not address how government bodies themselves should govern the internal use of AI simulation as a decision-support tool for policy design and evaluation. This is a distinct use case with its own risk profile — the accountability question is not whether an agent took an incorrect action, but whether and how simulation outputs should inform policy decisions as a form of ‘evidence’ or ‘data’. This thesis aims to provide governance guidelines specific to the use of LLM-based policy simulation within government, building on Singapore's existing governance architecture.

Methodology

I propose a mixed-methods design structured around three phases. LLM-based multi-agent simulation serves as the primary data collection method, producing qualitative and quantitative outputs that I systematically compare against documented real-world policy outcomes. The fine-tuning work constitutes a methodological development component. The framework and governance guidelines emerge from synthesis across all phases. Each phase could produce independently publishable findings while contributing to the integrated thesis.

Phase 1: Retrospective Simulation Validation (Year 1)

The first phase establishes the empirical foundation by studying the extent to which LLM-based simulation can reproduce known implementation dynamics and outcomes from past policy rollouts. Primary cases come from Singapore’s education policy context. Candidate cases could include the PSLE scoring reform and Full Subject-Based Banding initiatives, both of which generated well-documented stakeholder responses. Alternative cases from other Singapore policy domains will be considered if ethics review flags conflicts of interest arising from my professional role at MOE. An Australian comparator case — potentially drawn from education or social policy reforms — would provide cross-contextual validation and align with Crawford’s regional orientation; specific case selection would be confirmed with supervisors during the first semester.

The simulation procedure involves uploading policy documents as seed material to a GABM pipeline, configuring agent personas representing key stakeholder groups (teachers, school leaders, parents across varied socioeconomic profiles, students, policy officers), and running simulation rounds that generate implementation dynamics, resistance patterns, opinion shifts, and emergent workarounds. I then compare these outputs against documented outcomes from publicly available sources such as announced policy outcomes, parliamentary debates, media coverage, and academic evaluations (if available). Validity is assessed across dimensions of face validity (do simulated responses resemble documented stakeholder reactions?), construct validity (do agents exhibit theoretically expected behaviours?), and predictive validity (would the simulation have surfaced the implementation frictions that actually emerged?).

Phase 2: Fine-Tuning a Local Simulation Model (Year 2)

This phase develops a replicable methodology for locally grounding LLM-based policy simulation. The fine-tuning corpus draws on publicly available materials: Singapore educational policy documents, ministerial speeches, parliamentary debates, community feedback, and academic literature on Singapore education policy. Where ethics approval permits, semi-structured interviews with former teachers, parents, and school leaders could supplement the corpus — framed as retrospective perspectives on past policy rollouts rather than commentary on current policies.

The thesis investigates multiple approaches to local grounding, including LoRA fine-tuning of open-source base models, RAG-augmented prompting of frontier models such as Claude or GPT, and hybrid architectures where fine-tuned local models provide culturally grounded persona behaviour while larger models handle simulation orchestration. The full methodology is documented for replication, and the product would be a Singapore Policy Simulation Model (v1), released as an open-source resource.

Phase 3: Simulating Policy Evaluations and Guidelines Development (Year 3)

The third phase extends the simulation from policy implementation to policy evaluation. Using the implementation scenarios generated in Phases 1 and 2, I simulate a range of evaluation designs — from static approaches such as RCTs and quasi-experimental designs to more iterative and responsive approaches such as realist and developmental evaluation — to test their appropriateness under different implementation conditions. The underlying logic is that the nature of the policy should shape the evaluation design: a policy designed for uniform, high-fidelity implementation is more amenable to a static experimental design, while a policy designed for frontline autonomy calls for evaluation approaches that are responsive, iterative, and open to emergent outcomes. Simulation allows us to explore this matching before real-world evaluation resources are committed. The claim is carefully bounded: simulation does not replace evaluation; it improves evaluation design by reducing the risk of discovering implementation failure midway through a costly study.

The findings from this phase inform where simulation sits within the evaluation design process — whether it precedes, accompanies, or iterates with theory of change / logic model development — and specifies conditions for trusting simulation outputs. Where feasible, this framework will be refined through expert feedback from evaluation practitioners in the Singapore public sector and Australian government evaluation units.

Finally, drawing on learning across all three phases, this thesis produces governance guidelines addressing how LLM-based simulation should be used as a decision-support tool in public sector policy design and evaluation, building on Singapore's existing Model AI Governance Frameworks for Generative AI and Agentic AI.

Timetable

The thesis follows a full-time three-year candidature.

Period	Milestone
Year 1, Sem 1	Coursework; literature review across all clusters; ethics application; configure simulation environment
Year 1, Sem 2	Phase 1: Simulation runs on Singapore cases; validity assessment; Australian comparator case selection confirmed with supervisors
Year 1 End	Research proposal seminar; Paper 1 drafted (validation study)
Year 2, Sem 1	Phase 2: Fine-tuning corpus assembled; LoRA fine-tuning and comparative validation against Phase 1 baseline
Year 2, Sem 2	Paper 2 drafted (fine-tuning methodology); Phase 3 data collection begins
Year 2 End	Paper 1 submitted; Singapore Policy Simulation Model (v1) released
Year 3, Sem 1	Phase 3: Evaluation simulation and framework development; expert feedback where feasible; governance guidelines drafted; Paper 3 drafted
Year 3, Sem 2	Thesis integration, revision, and submission; Papers 2 and 3 submitted

Resources

The main resource required is computing — both hardware and API credits. The minimum hardware requirement is an Apple Silicon Mac with 32GB RAM running Ollama and LM Studio, sufficient for LoRA fine-tuning of 7B-parameter models locally. API costs for larger simulation runs on platforms such as MiroFish are estimated at SGD 500–2,000 per run. Cloud computing for fine-tuning larger models, if needed, is estimated at AUD 2,000–5,000 over the candidature.

For fieldwork, expert interviews in Singapore will be conducted primarily remotely or during home visits. I would welcome advice on possible local visits and interviews in Canberra to support the Australian comparator case.

The simulation components of the thesis will benefit from an informal technical advisor — ideally from ANU's College of Engineering and Computer Science or the broader CAMEL-AI/AgentTorch research community. This is a resource need, not a supervision gap: the supervisors provide the disciplinary expertise the thesis requires, but I would benefit from a sounding board on the technical aspects of simulation and fine-tuning.

I am awarded a Ministry of Education (Singapore) Postgraduate Scholarship for this PhD application, covering three years of candidature.

Feasibility and Limitations

The methodology is framed as generic LLM-ABM, not as a study of any single platform, since these platforms and technologies are advancing rapidly. As of writing, MiroFish appears to be the ideal primary simulation tool, but OASIS, AgentTorch, or newer simulation tools could substitute without altering the research design or theoretical contribution. Second, the well-documented Western bias in LLM training data is not a confound to manage but the research problem itself. Phase 2 investigates it empirically through fine-tuning, which would require local validation. Third, I am not a machine learning expert and have developing (and improving) experience with LLM fine-tuning and deployment. I manage this by treating fine-tuning as a methodology chapter rather than a results chapter, by seeking an informal technical advisor, and by maintaining throughout that the contribution is to public policy and policy evaluation, not to machine learning. Finally, my professional role at the Ministry of Education (Singapore) would require me to restrict Singapore data to publicly available archival sources and position this thesis as contributing to governance practice broadly rather than to the Ministry of Education (Singapore) specifically.

Institutional Fit

Crawford POGO is ideal for this thesis because its interdisciplinary tradition provides the breadth required to bridge policy design and implementation, evaluation methodology, computational simulation methods, and AI governance. Second, Crawford's Asia-Pacific focus makes Singapore and its neighbouring governance systems relevant. And third, POGO's openness to methodological innovation aligns with the thesis's aim to study generative agent-based modelling simulations for public policy implementation and policy evaluation.

I propose Associate Professor Michael Di Francesco as primary supervisor. His expertise in public administration reform and policy implementation provides the empirical anchor for a thesis concerned with why implementation dynamics diverge from policy intent. As secondary supervisor, I propose Associate Professor Mark Chou, whose political theory background is essential for the governance and normative dimensions—particularly the question of what it means to simulate publics for governance purposes and the ethical conditions under which such simulation can be used responsibly.

Bibliography

- Axelrod, R. (1997). *The complexity of cooperation: Agent-based models of competition and collaboration*. Princeton University Press.
- Chong, W. H., & Lee, B. O. (2021). Understanding effective implementation of prevention education programmes: Perspective from Singapore schools. *Asia-Pacific Education Researcher*, 30(1), 23–32.
- Epstein, J. M. (2006). *Generative social science: Studies in agent-based computational modeling*. Princeton University Press.
- Funnell, S., & Rogers, P. (2011). *Purposeful program theory: Effective use of theories of change and logic models*. Jossey-Bass.
- Ghaffarzadegan, N., Majumdar, A., Williams, R., & Hosseinichimeh, N. (2024). Generative agent-based modeling: An introduction and tutorial. *System Dynamics Review*, 40(1), 1–29.
- Gopinathan, S. S., & Lim, E. (2021). Reform implementation lessons: A case study of high-performing Singapore. In C. McLaughlin & A. Ruby (Eds.), *Implementing educational reform* (pp. 149–170). Cambridge University Press.

- Gao, C., Lan, X., Li, N., Yuan, Y., Ding, J., Zhou, Z., Xu, F., & Li, Y. (2024). Large language models empowered agent-based modeling and simulation: A survey and perspectives. *Humanities and Social Sciences Communications*, 11, 1–16.
- Hu, E., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2021). LoRA: Low-rank adaptation of large language models. *arXiv:2106.09685v2*.
- Infocomm Media Development Authority (IMDA) & AI Verify Foundation. (2024). *Model AI Governance Framework for Generative AI*.
- Infocomm Media Development Authority (IMDA). (2026). *Model AI Governance Framework for Agentic AI* (Version 1.0).
- Kovač, G., Sawayama, M., Portelas, R., Colas, C., Dominey, P. F., & Oudeyer, P.-Y. (2023). Large language models as superpositions of cultural perspectives: The unexpected perspective shift effect – A pitfall of using psychological tools for large language models. *arXiv:2307.07870*.
- Li, Y., Das, S., & Shirado, H. (2025). What makes LLM agent simulations useful for policy? Insights from an iterative design engagement in emergency preparedness. *arXiv:2509.21868*.
- Lipsky, M. (2010). *Street-level bureaucracy: Dilemmas of the individual in public services* (2nd ed.). Russell Sage Foundation.
- Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *Proceedings of UIST 2023*.
- Patton, M. Q. (2010). *Developmental evaluation: Applying complexity concepts to enhance innovation and use*. Guilford Press.
- Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose opinions do language models reflect? *Proceedings of the 40th International Conference on Machine Learning, PMLR 202*, 1-34.
- Spillane, J. P. (2006). *Distributed leadership*. Jossey-Bass.
- Larooij, M., & Törnberg, P. (2026). Validation is the central challenge for generative social simulation: A critical review of LLMs in agent-based modeling. *Artificial Intelligence Review*, 59(15), 1–31.
- Trinidad, J. E. (2024). An organizational sociology of education: Using structural, network, and ecological perspectives to study schools. *Sociological Inquiry*, 94(4), 968–993.
- Viennet, R., & Pont, B. (2017). Education policy implementation: A literature review and proposed framework. *OECD Education Working Papers*, 162.